SOFTWARE ENGINEERING

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LAB : 6

Lab 7 – Fuzzing for Input Validation Bugs.

SS1 – Execution of Original Buggy Application

```
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7> python house_painting_quote_parser.py
==== Running clean_colour_name ====
Enter a paint colour name (e.g. Ocean Blue): Ocean Blue
Colour code: ocean_blue

==== Running room_area ====
Enter room size as LENGTHxWIDTH in feet (e.g. 12x10): 12x10
Room area (sq ft): 120

==== Running total_wall_area ====
Enter wall areas separated by ; (e.g. 120;95;110): 120;95;110
Total wall area (sq ft): 325
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7>
```

```
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7> python -m pytest test_house_painting_quote_parser.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.6.0
rootdir: C:\Users\91944\L6_L7_63_house_painting_quote\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.14
collected 3 items

test_house_painting_quote_parser.py ...                                  [100%]

============================== 3 passed in 4.82s ==============================
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7>
```

SS2 – Identification of Bugs Using Hypothesis Fuzz Testing

```
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7> python -m pytest test_house_painting_quote_parser.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.6.0
rootdir: C:\Users\91944\L6_L7_63_house_painting_quote\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.14
collected 3 items

test_house_painting_quote_parser.py FFF                                  [100%]

================================== FAILURES ===================================
_____________________ test_clean_colour_name_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_clean_colour_name_no_crash(s):
        try:
>           clean_colour_name(s)
```

```
=========================== short test summary info ===========================
FAILED test_house_painting_quote_parser.py::test_clean_colour_name_no_crash - AssertionError: clean_colour_name crashed with: 'NoneTy
pe' object has no attribute 'strip'
FAILED test_house_painting_quote_parser.py::test_room_area_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-ex
ceptions)
FAILED test_house_painting_quote_parser.py::test_total_wall_area_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2
sub-exceptions)
============================== 3 failed in 2.41s ==============================
```

SS3 – Execution of Fixed Application

```
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7> python house_painting_quote_parser.py
==== Running clean_colour_name ====
Enter a paint colour name (e.g. Ocean Blue): Ocean Blue
Colour code: ocean_blue

==== Running room_area ====
Enter room size as LENGTHxWIDTH in feet (e.g. 12x10): 12x10
Room area (sq ft): 120

==== Running total_wall_area ====
Enter wall areas separated by ; (e.g. 120;95;110): 120;95;110
Total wall area (sq ft): 325
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7>
```

SS4 – Successful Execution of Hypothesis Tests

```
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7> python -m pytest test_house_painting_quote_parser.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.6.0
rootdir: C:\Users\91944\L6_L7_63_house_painting_quote\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.14
collected 3 items

test_house_painting_quote_parser.py ...                                                                                            [100%]

============================================================= 3 passed in 1.35s =============================================================
PS C:\Users\91944\L6_L7_63_house_painting_quote\LAB7>
```